Part V

Constrained Optimization I

Lecture Outline

Constrained Optimization

In this lecture we will discuss C^2 constrained optimization problems, i.e., we consider

$$\min_{x \in R} f(x)$$

where the C^2 cost (or objective) function

$$f : R \subset \mathbb{R}^n \to \mathbb{R};\ x \mapsto f(x),$$

and the C^2 equality and inequality constraint functions

$$\mathbf{a} : \mathbb{R}^n \to \mathbb{R}^p;\ x \mapsto \mathbf{a}(x) = [(a_1(x) \cdots a_p(x)]^\top,$$
$$\mathbf{c} : \mathbb{R}^n \to \mathbb{R}^q;\ x \mapsto \mathbf{c}(x) = [c_1(x) \cdots c_q(x)]^\top,$$

are given, and the feasible region R is

$$R = \{x \in \mathbb{R}^n \mid \mathbf{a}(x) = 0 \text{ and } \mathbf{c}(x) \geq_e 0\},$$

with $\mathbf{c}(x) \geq_e 0$ defined by $c_i(x) \geq 0,\ i = 1, \ldots, q$ (the subscript $_e$ stand for entry-wise)

The above will be written compactly as

$$\min f(x) \quad s.t \quad \mathbf{a}(x) = 0,\ \mathbf{c}(x) \geq_e 0.$$

Example

Linear programming (LP):

$$\min c^\top x \quad s.t \quad Ax = b,\ x \geq_e 0$$

LP (nonstandard):

$$\min \alpha^\top z \quad s.t \quad Bz \geq_e 0$$

Quadratic programming (QP):

$$\min x^\top Qx + b^\top x + c \quad s.t \quad Ax = b,\ Cx \geq_e 0$$

with $Q \geq 0$

Convex programming (CP):

$$\min f(x) \quad s.t \quad Ax = b,\ \mathbf{c}(x) \geq_e 0$$

with $f(x)$ and $-\mathbf{c}(x)$ convex functions.

Technical remark: Not all convex sets can be expressed as the solution to $Ax = b,\ \mathbf{c}(x) \geq_e 0$ e.g., the convex hull of a fractal set (such as the Koch snowflake)

Definition

A point x' satisfying the equality constraint $\mathbf{a}(x) = 0$ is called regular if the Jacobian of the equality constraint function $\mathbf{a}$ at x'

$$J_{\mathbf{a}}(x') = [\nabla a_1(x') \cdots \nabla a_p(x')]^\top \in \mathbb{R}^{p \times n},$$

has full row rank, or equivalently, if $\nabla a_1(x'), \ldots, \nabla a_p(x')$ are linearly independent.

Note that a necessary condition for x' to be regular is that

$$p \leq n,$$

i.e., regularity requires that there are at most n equality constraints.

Regular points

Let $n = 3$, $p = 2$ and

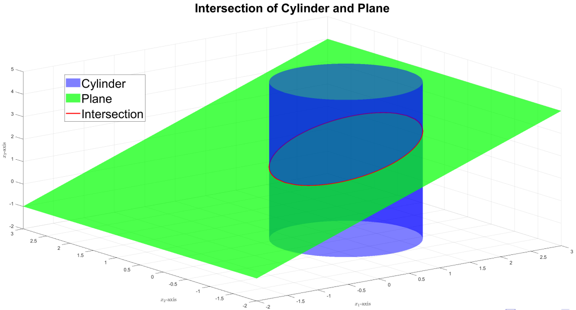
$$a_1(x) = -x_1 + x_3 - 1, \qquad a_2(x) = x_1^2 + x_2^2 - 2x_1.$$

Hence

$$\mathbf{a}(x) = \begin{bmatrix} -x_1 + x_3 - 1 \\ x_1^2 + x_2^2 - 2x_1 \end{bmatrix},$$

$$J_{\mathbf{a}}(x) = \begin{bmatrix} -1 & 0 & 1 \\ 2x_1 - 2 & 2x_2 & 0 \end{bmatrix} \sim \begin{bmatrix} -1 & 0 & 1 \\ 0 & 2x_2 & 2x_1 - 2 \end{bmatrix}$$

So $J_{\mathbf{a}}(x)$ has full row rank except at points $\bar{x} = (1, 0, x_3)$. But $a_2(\bar{x}) = -1$ implying that all points satisfying the constraints are regular.



Regular points for linear equality constraints

Assume affine linear equality constraint function, that is,

$$\mathbf{a}(x) = Ax - b,$$

where $A \in \mathbb{R}^{p \times n}$ and $b \in \mathbb{R}^p$. Hence the equality constraints $\mathbf{a}(x) = 0$ can be written as linear equality constraints

$$Ax = b.$$

In this case we note the equality constraints define an affine linear subspace in $\mathbb{R}^n$, and that

$$J_{\mathbf{a}}(x) = A,$$

so x is regular if

$$\operatorname{rank}(A) = p,$$

in which case we say that the constraints are consistent without redundancy.

Regular points for linear equality constraints

$\boxed{\text{Points } x \text{ satisfying the linear equality constraint } Ax = b \text{ are regular if } rank(A) = p.}$

If $rank(A) < p$ then we need to consider two cases:

- Inconsistent constraints: $rank([A\ b]) > \text{rank}(A)$.
- Consistent constraints with redundancy: $rank([A\ b]) = rank(A)$.

Regular points for linear equality constraints

Points x satisfying the linear equality constraint $Ax = b$ are regular if $rank(A) = p$.

If $rank(A) < p$ then we need to consider two cases:

- Inconsistent constraints: $rank([A\ b]) > \text{rank}(A)$.
- Consistent constraints with redundancy: $rank([A\ b]) = rank(A)$.

Inconsistent constraints means that b is not contained in range(A), so $Ax = b$ has no solution. That is, the equality constraints contain a contradiction which must be eliminated before proceeding with any future optimization.

Example

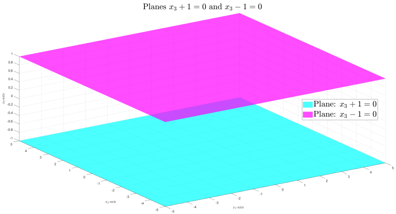
Let $n = 3$, $p = 2$ and

$$a_1(x) = x_3 + 1, \qquad a_2(x) = x_3 - 1.$$

Hence

$$\mathbf{a}(x) = Ax - b = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} x - \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

So $\text{rank}([A\ b]) = 2 > 1 = \text{rank}(A)$, implying that the constraints are inconsistent.



Regular points for linear equality constraints

Points x satisfying the linear equality constraint $Ax = b$ are regular if $rank(A) = p$.

If $rank(A) < p$ then we need to consider two cases:

- Inconsistent constraints: $rank([A\ b]) > \text{rank}(A)$.
- Consistent constraints with redundancy: $rank([A\ b]) = rank(A)$.

Consistent constraints with redundancy means that $b \in range(A)$. In this case, we may reduce the p constraints $Ax = b$ to an equivalent set of $p' = \text{rank}(A)$ equality constraints by removing some of the constraints[1]

Example

Let $n = p = 3$ and

$$a_1(x) = x_1 + x_2 + 2x_3 - 2, \quad a_2(x) = x_1 + x_2 + 2x_3 - 2, \quad a_3(x) = x_1 - x_2 + 2.$$

Hence

$$\mathbf{a}(x) = Ax + b = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 1 & 2 \\ 1 & -1 & 0 \end{bmatrix} x - \begin{bmatrix} 2 \\ 2 \\ -2 \end{bmatrix}.$$

with $rank([A\ b]) = 2 = rank(A)$.

[1] Technical remark: A methodical way is to use a reduced SVD of A.

Regular points and active constraints

Definition

An inequality constraint $c_i(x) \geq 0$ is said to be active at $\bar{x}$ if $c_i(\bar{x}) = 0$ and inactive at $\bar{x}$ if $c_i(\bar{x}) > 0$.

A point x' satisfying the constraints $\mathbf{a}(x) = 0$ and $\mathbf{c}(x) \geq_e 0$ is called regular (for the active constraints) if

$$\nabla a_1(x'), \dots, \nabla a_p(x'), \nabla c_j(x'),\ j \in K$$

are linearly independent, with $K = K(x')$ the set of indices of active constraints at x'.

Note that K might be empty, and that the concept of active constraints is (numerical) important since it dictate which inequality constraints must be taken into account in a neighborhood of a point.

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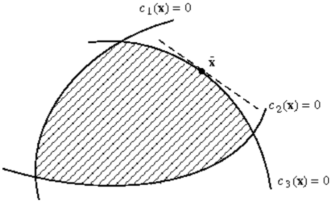

Figure: Active and inactive constraints.

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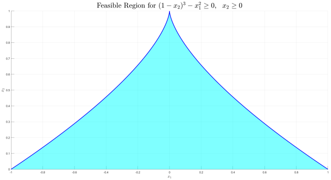
Example

The point $x' = [0\ 1]^\top$ is not a regular point for the constraints:

$$(1 - x_2)^3 - x_1^2 \geq 0, \quad x_2 \geq 0$$



Linear inequality constraints

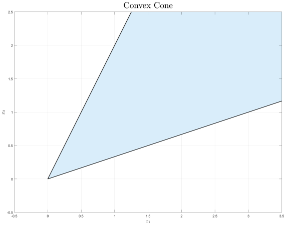
Assume affine linear inequality constraint functions, that is,

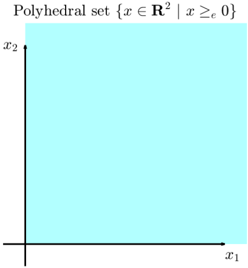
$$\mathbf{c}(x) = Cx - d,$$

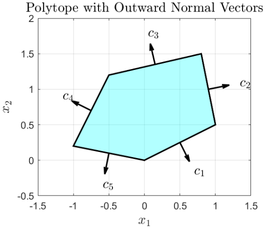
where $C \in \mathbb{R}^{q \times n}$ and $d \in \mathbb{R}^q$. Hence the inequality constraints $\mathbf{c}(x) \geq_e 0$ can be written as linear inequality constraints

$$Cx \geq_e d.$$

In this case, we note that the inequality constraints define a convex set in $\mathbb{R}^n$ (or more precisely, a convex polyhedron in $\mathbb{R}^n$).







Inequality to equality constraints

Consider

$$\min f(x) \quad s.t \quad \mathbf{a}(x) = 0, \ \mathbf{c}(x) \geq_e 0.$$

By introducing (slack) variables $y = [y_1 \cdots y_q]^\top \in \mathbb{R}^q$ we may rewrite this optimization problem as

$$\min h(z) \quad s.t \quad \mathbf{b}(z) = 0,$$

where

$$\begin{aligned} z &= [x \ y]^\top, \\ h(z) &= f(x), \\ \mathbf{b}(z) &= [a_1(x) \cdots a_p(x) \ \alpha_1(z) \cdots \alpha_q(z)]^\top, \\ &\alpha_i(z) = c_i(x) - y_i^2 \end{aligned}$$

That is, by introducing slack variables we have transformed an optimization problem with equality and inequality constraints into an equivalent optimization problem with only equality constraints.

Equality to inequality constraints

Consider

$$\min f(x) \quad s.t \quad \mathbf{a}(x) = 0, \ \mathbf{c}(x) \geq_e 0.$$

We can also rewrite this optimization problem to one with only inequality constraints as follows

$$\min f(x) \quad s.t \quad \mathbf{b}(x) \geq_e 0,$$

where

$$\mathbf{b}(z) = [c_1(x) \cdots c_q(x) \ a_1(x) \cdots a_p(x) \ \ - a_1(x) \cdots - a_p(x)]^\top$$